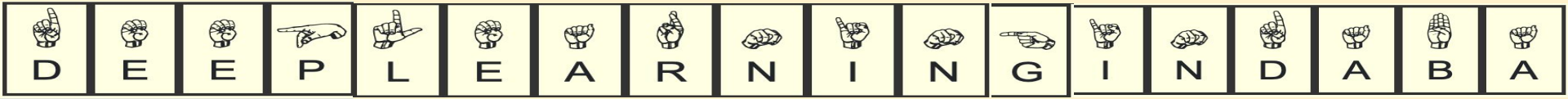

AFRICAN SIGN
LANGUAGES
TRANSLATION
Shester MSOUOBU
Mardiyyah ODUWOLE
Steven KOLAWOLE
DEEP
LEARNING
INDABA





source: WHO Word Report Hearing 2021

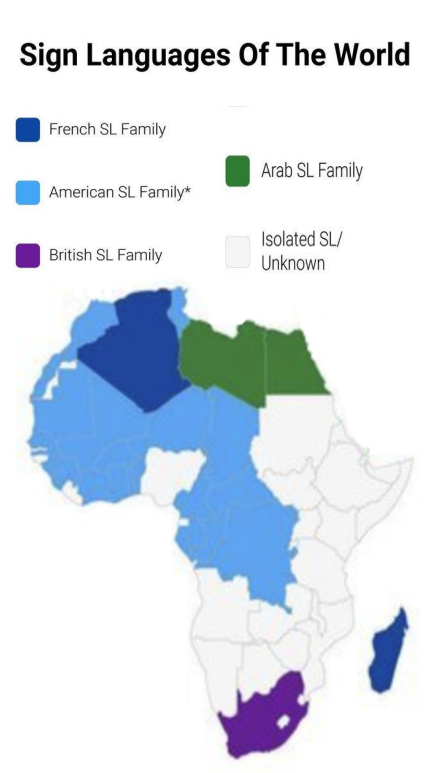




- **43 different sign languages, 13 of which have a foreign origin.**
- **136 million deaf or hard of hearing people in Africa.**
- **Sign language is the most efficient way to communicate with people living with hearing disability.**

source: WHO Word Report Hearing 2021


Timnit Gebru @timnitGebru · Aug 10
Again.
Haben Girma @HabenGirma · Aug 9
Hi, I'm a #Deafblind person strongly recommending #AI researchers & others in #tech talk to Deafblind people rather than making assumptions about us. This piece by @emilymbender explains why it's harmful to compare AI to Deafblind people:
medium.com/@emilymenonben...
1
14
58



Local Sign Languages:

- Adamorobe Sign Language used in Eastern Region of Ghana

- Nanabin Sign Language used in Central Region of Ghana.

- Bura Sign Language which is used in the south-east of Biu, Nigeria.



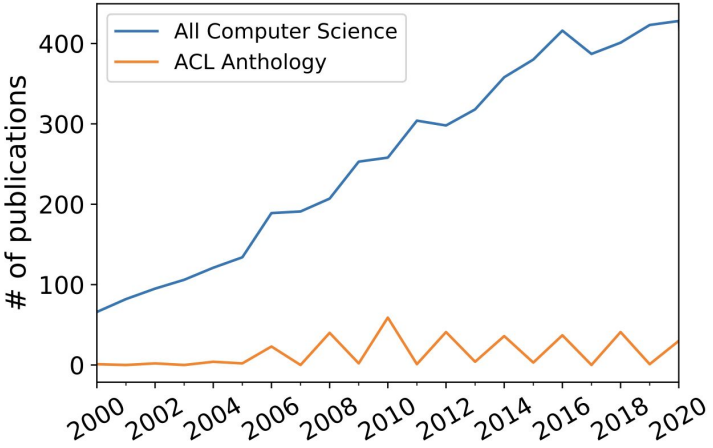
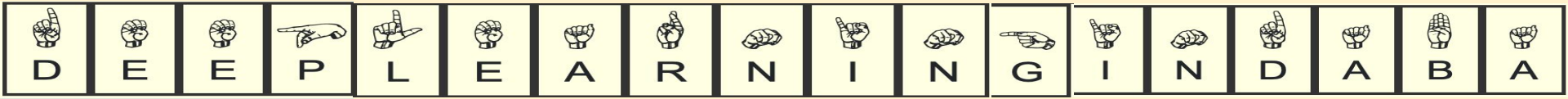


Figure 1: Evolution of the number of publications referring to sign language in its title from computer science venues and in the ACL anthology. Publications in computer science are extracted from the Semantic Scholar

- Signed languages exhibit all the fundamental linguistic properties of natural language, we believe that tools and theories of Natural Language Processing (NLP) are crucial towards its modeling.
- Recent advances in Video Processing and Machine Translation Techniques.
- Activeness of the African Community towards including deaf people in the society. TV News Interpreters, Music Videos translated into Sign Language, Sign Language Courses.
- We are in collaboration with deaf people , sign language university instructors and accredited sign language TV interpreters.
- Most African Sign Languages are derived from non-African Sign Languages.

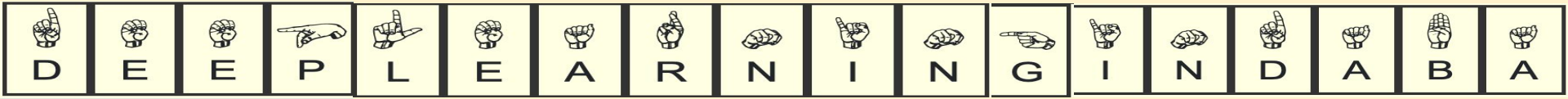






- Lack of African Sign Language Datasets for Sign Language Translation
- Lack of Models trained on African Sign Languages
- Little or no publication including African Sign Languages.
- Signed languages challenges for NLP ;
  *visual-gestural modality,
  *simultaneity,
  *spatial coherence,
  *lack of written form.

Mr. Marco Stanley Nyarko
Sign Language Lecturer at KNUST





DOOR INTERNATIONAL
DEAF REACHING DEAF FOR CHRIST

- 7 African Sign Languages

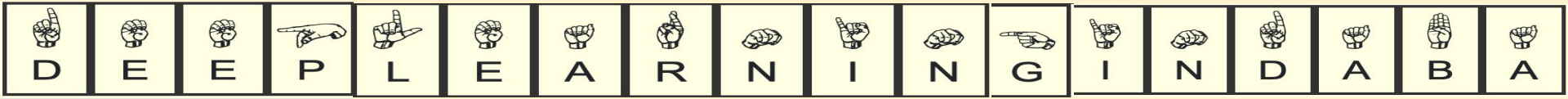
Nigeria🇳🇬, Ghana🇬🇭, Kenya🇰🇪, Mozambique🇲🇿
Ethiopia🇪🇹, Burundi🇧🇮, South Sudan🇸🇸

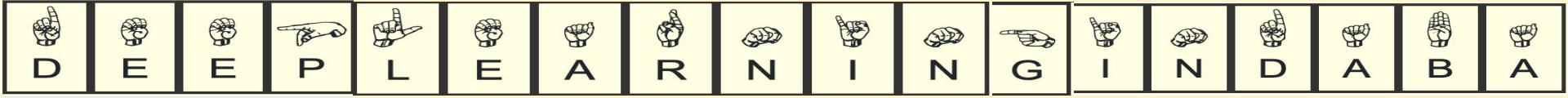
- 15 African Sign Languages

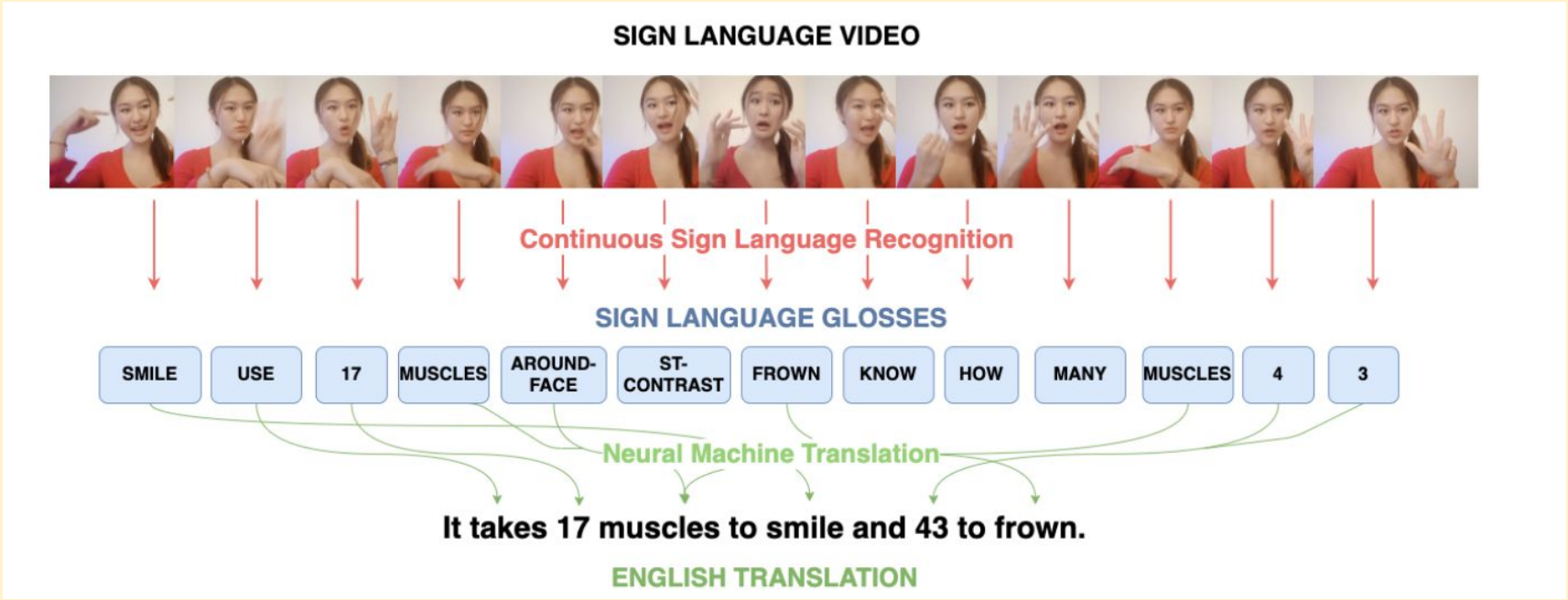
Nigeria🇳🇬, Ghana🇬🇭, Kenya🇰🇪, Mozambique🇲🇿,
Ethiopia🇪🇹 Burundi,🇧🇮 Uganda 🇺🇬, Angola🇦🇴,
Congo🇨🇬, Malawi🇲🇼, Zambia🇿🇲, Zimbabwe🇿🇼,
Rwanda🇷🇼, Ivory Coast🇨🇮, Tanzania🇹🇿

JW LIBRARY

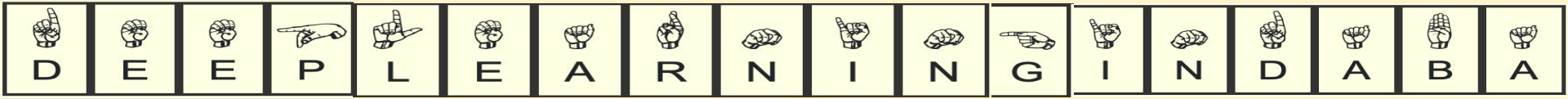
JW Library Sign Language APK

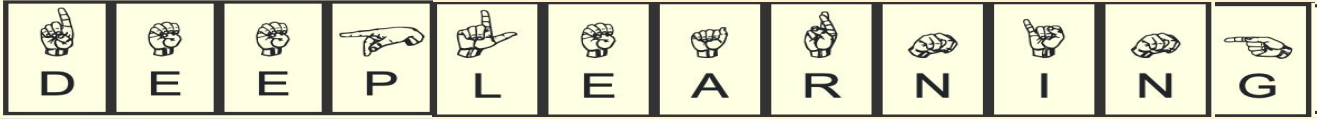


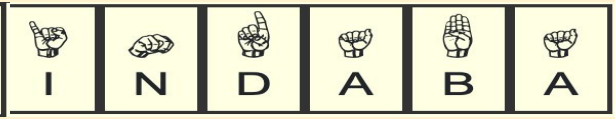




Credit: Yin, K. and Read, J., 2020, December. Better sign language translation with STMC-transformer. In *Proceedings of the 28th International Conference on Computational Linguistics* (pp. 5975-5989).







Small Data? No Problem! Exploring the Viability of Pretrained Multilingual Language Models for Low-Resource Languages

Kelechi Ogueji, Yuxin Zhu, and Jimmy Lin

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University of Waterloo

{kelechi.ogueji, yuxin.zhu, jimmylin}@uwaterloo.ca

Abstract

Pretrained multilingual language models have been shown to work well on many languages for a variety of downstream NLP tasks. However, these models are known to require a lot of training data. This consequently leaves out a huge percentage of the world's languages as they are under-resourced. Furthermore, a major motivation behind these models is that lower-resource languages benefit from joint training with higher-resource languages. In this work, we challenge this assumption and present the first attempt at training a multilingual language model on only low-resource languages. We show that it is possible to train competitive multilingual language models on less than 1 GB of text. Our model, named AfriBERTa, covers 11 African languages, including the first language model for 4 of these languages. Evaluations on named entity recognition (NER) and text classification downstream tasks on 10 low-resource languages. Our models outperform larger models like mBERT and XLM-R by up to 10 F1 points on text classification, and also outperform that they help boost transfer to lower-resource languages. Hence, there has been no known attempt to investigate if it is possible to pretrain multilingual language models solely on low-resource languages without any transfer from higher-resource languages, despite the numerous benefits that this could provide. Motivated by this gap in the literature, the goal of our work is to explore the viability of multilingual language models pretrained from scratch on low-resource languages and to understand how to pretrain such models in this setting.

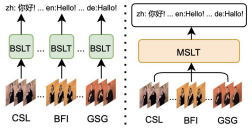
We introduce AfriBERTa, a transformer-based multilingual language models trained on 11 African languages, all of which are low-resource.[1] We evaluate this model on named entity recognition (NER) and text classification downstream tasks on 10 low-resource languages. Our models outperform larger models like mBERT and XLM-R by up to 10 F1 points on text classification, and also outperform

MLSLT: Towards Multilingual Sign Language Translation

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Abstract

Most of the research to date focuses on bilingual sign language translation (BSLT). However, such models are inefficient in building multilingual sign language translation systems. To solve this problem, we introduce the multilingual sign language translation (MSLT) task. It aims to use a single model to complete the translation between multiple sign languages and spoken languages. Then, we propose MLSLT, the first MSLT model, which contains two novel dynamic routing mechanisms for controlling the degree of parameter sharing between different languages. Intra-layer language-specific routing controls the proportion of data flowing through shared parameters and language-specific parameters from the token level through a soft gate within the layer, and inter-layer language-specific routing controls and learns the data flow path of different languages at the language level through a soft gate between layers. In order to evaluate the performance of MLSLT, we collect the first publicly available multilingual sign language understanding dataset, Spreadthesign-Ten (SP-10), which contains up



(a) MSLT system based on multi-models (b) MSLT system based on single model

Figure 1. An example to illustrate the advantages of the MSLT model over the BSLT model in constructing a multilingual sign language translation system.

motivates researchers to design more efficient and accurate sign language translation systems [23, 24, 32, 56].

Due to the lack of data, almost all previous studies focus on building a bilingual sign language translation (BSLT)

Better Sign Language Translation with STMC-Transformer

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Institut Polytechnique de Paris
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Abstract

Sign Language Translation (SLT) first uses a Sign Language Recognition (SLR) system to extract sign language glosses from videos. Then, a translation system generates spoken language translations from the sign language glosses. This paper focuses on the translation system and introduces the STMC-Transformer which improves on the current state-of-the-art by over 5 and 7 BLEU respectively on gloss-to-text and video-to-text translation of the PHOENIX-Weather 2014T dataset. On the ASLG-PC12 corpus, we report an increase of over 16 BLEU.

2870v1 [cs.CV] 25 May 2022

Open-Domain Sign Language Translation Learned from Online Video

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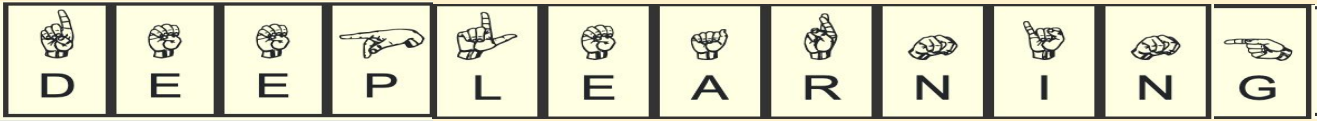
Abstract

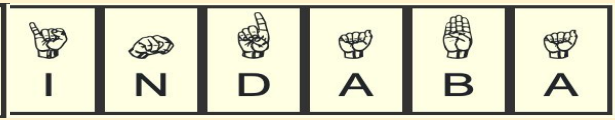
Existing work on sign language translation—that is, translation from sign language videos into sentences in a written language—has focused mainly on (1) data collected in a controlled environment or (2) data in a specific domain, which limits the applicability to real-world settings. In this paper, we introduce OpenASL, a large-scale ASL-English dataset collected from online video sites (e.g., YouTube). OpenASL contains 288 hours of ASL videos in various domains (news, VLOGs, etc.) from over 200 signers and is the largest publicly available ASL translation dataset to date. To tackle the challenges of sign language translation in realistic settings and without glosses, we propose a set of techniques including sign search as a pretext task for pre-training and fusion of mouthing and handshape features. The proposed techniques produce consistent and large improvements in translation quality, over baseline models based on prior work.[1]

and spoken languages. Interest in sign language research has recently been growing in the computer vision (CV) (Bragg et al., 2019) and natural language processing (NLP) communities (Yin et al., 2021b)

In this paper, we study sign language translation (SLT),[3] the task of translating continuous signing video into written language sentences. Unlike other sign language processing tasks such as sign spotting (Buehler et al., 2009) or continuous sign language recognition (Dreuw et al., 2007), SLT has not been studied until recently (Camgoz et al., 2018) and is still restricted to specific domains (e.g., weather forecasts (Camgoz et al., 2018), emergency situations (Ko et al., 2019)), characterized by small vocabulary size and lack of visual variability. Lack of large-scale translation datasets in the wild is the central challenge in developing SLT technologies serving real-world use cases.

In terms of translation modeling, existing SLT approaches (Camgoz et al., 2018, 2020a,a; Zhou et al., 2021; Yin and Read, 2020; Cao et al., 2021;





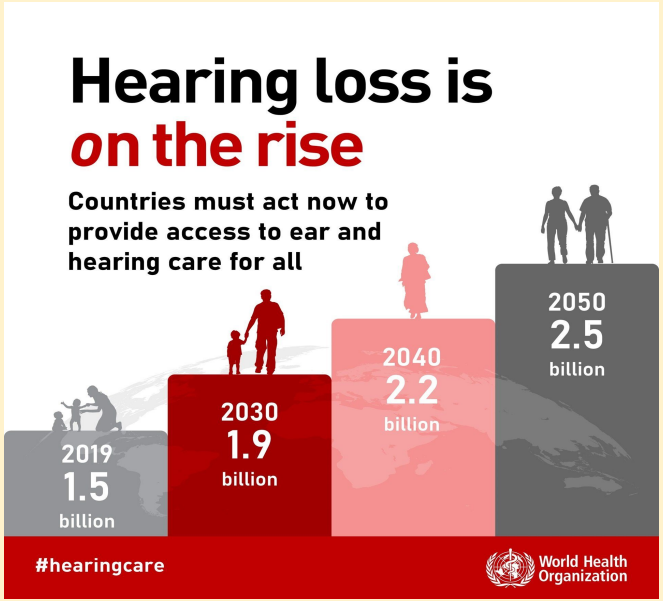


- Real-time automatic sign language interpretations especially in domains such as health.
- Better documentation of Endangered African sign languages
- Educational tools for sign language learners
- Tools for query and retrieval of information from signed language videos
- Personal assistants that react to signed languages

source: http://www.ayelefoundation.org/gsl/

DEEPLEARNING INDABA

WHY NOW ?



source: WHO Word Report Hearing 2021

- Lots of ongoing work on non-African Sign African Sign Language
- 80% of people with hearing loss live in middle and low level. We believe we need to be at the forefront to create push for technology.



Deep Learning Indaba , Tunisia 2022, IDEATHON